AC6925E Datasheet

Zhuhai Jieli Technology Co.,LTD

Version：V1.0

Date：2018.09.05

ZHUHAI JIELI TECHNOLOGY

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AC6925E Features

High performance 32-bit RISC CPU

- RISC 32-bit CPU
- DC-160MHz operation
- Support DSP instructions
- 64Vectored interrupts
- 4 Levels interrupt priority

Flexible I/O

- 12 GPIO pins
- All GPIO pins can be programmable as input or output individually
- All GPIO pins are internal pull-up/pull-down selectable individually
- CMOS/TTL level Schmitt triggered input
- External wake up/interrupt on all GPIOs

Peripheral Feature

- One full speed USB 2.0 OTG controller
- Four multi-function 16-bit timers, support capture and PWM mode
- Three 16-bit PWM generator for motor driving
- One full-duplex basic UART
- Two full-duplex advanced UART
- One SPI interface supports host and device mode
- One SD Card Host controller
- One IIC interface supports host and device mode
- One Quadrate decoder
- Watchdog
- 1 Crystal Oscillator
- 16-bit Stereo DAC with headphone amplifier, SNR >= 95dB
- 1 channel ADC , SNR >= 90dB
- 1 channel MIC amplifier
- 2 channels Stereo analog MUX
- 8 channels 10-bit ADC
- 2 channels 8 levels Low Voltage Detector
- Power-on reset
- Embedded PMU support low power mode

Bluetooth Feature

- CMOS single-chip fully-integrated radio and baseband
- Compliant with Bluetooth V5.0+BR+EDR+BLE specification
- Bluetooth Piconet and Scatternet support
- Meet class2 and class3 transmitting power requirement
- Support GFSK and π/4 DQPSK all paket types

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- Provides +2dbm transmitting power
- receiver with -89dBm sensitivity
- Support a2dp\avctp\avdtp\avrcp\hfp\spp\smp\att\gap\gatt\rfcomm\sdp\l2cap profile

FM Tuner

- Support worldwide frequency band 76-108MHz
- Fully integrated digital low-IF tuner & frequency synthesizer
- Autonomous search tuning
- Digital auto gain control (AGC)
- Digital adaptive noise cancellation
- Programmable de-emphasis (50/75 uS)
- Receive signal strength indicator (RSSI)
- Radio search in multi-channel simultaneously
- Digital volume control

Power Supply

- VBAT is 2.2V to 5.5V
- VDDIO is 2.2V to 3.6V

Packages

- QSOP24

Temperature

- Operating temperature: -20℃ to +70℃
- Storage temperature: -65℃ to +150℃

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1、Pin Definition

1.1 Pin Assignment

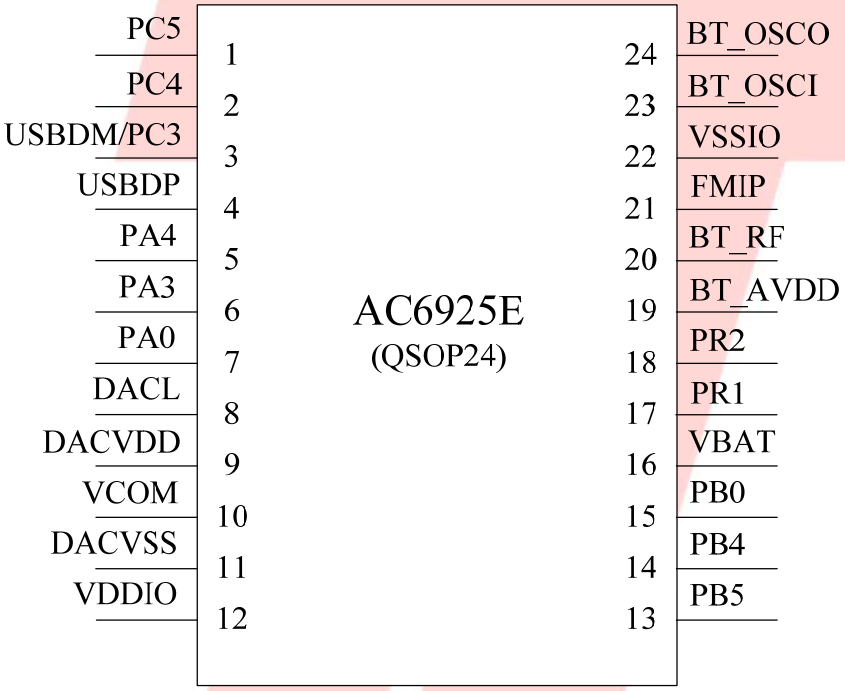


Figure 1-1 AC6925E_QSOP24 Package Diagram

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1.2 Pin Description

Table 1-1 AC6925E_QSOP24 Pin Description

PIN NO.	Name	I/O Type	High Drive (mA)	Function	Other Function
1	PC5	I/O	24	GPIO	SD1CLKA：SD1 Clock(A); SPI1DOB：SPI1 Data Out(B); UART2RXD：Uart2 Data In(B); IIC_SDA_B：IIC SDA(B);
2	PC4	I/O	24	GPIO	SD1CMDA：SD1 Command(A); SPI1CLKB：SPI1 Clock(B); UART2TXD：Uart2 Data Out(B); IIC_SCL_B：IIC SCL(B);
3	PC3	I/O	24	GPIO	SD1DAT0A：SD1 Data0(A); SPI1DIB：SPI1 Data In(B); UART0RXC：Uart0 Data In(C); TMR3：Timer3 Clock Input; ADC10：ADC Input Channel 10;
	USBDM	I/O	4	USB Negative Data (pull down)	UART1RXD：Uart1 Data In(D); SPI2DOB：SPI2 Data Out(B); IIC_SDA_A：IIC SDA(A); ADC11：ADC Input Channel 11;
4	USBDP	I/O	4	USB Positive Data (pull down)	UART1TXD：Uart1 Data Out(D); SPI2CLKB：SPI2 Clock(B); IIC_SCL_A：IIC SCL(A);
5	PA4	I/O	24	GPIO	PWM1：Timer1 PWM Output; AMUX1R：Simulator Channel1 Right; ADC1：ADC Input Channel 1; UART2RXA：Uart2 Data In(A); Touch11：Touch Input Channel 11;
6	PA3	I/O	24	GPIO	AMUX1L：Simulator Channel1 Left; ADC0：ADC Input Channel 0; UART2TXA：Uart2 Data Out(A); Touch10：Touch Input Channel 10;
7	PA0	I/O	24	GPIO	MIC：MIC Input Channel; UART0RXB：Uart0 Data In(B);
8	DACL	O	/	DAC Left Channel	
9	DACVDD	P	/	DAC Power	
10	VCOM	P	/	DAC Reference	
11	DACVSS	P	/	Ground	

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12	VDDIO	P	/	IO Power 3.3v	
13	PB5	I/O	8	GPIO	UART0TXB：Uart0 Data Out(B); AMUX0R：Simulator Channel0 Right; SPI1DOA：SPI1 Data Out(A); SD0CLKB：SD0 Clock(B); ADC9：ADC Input Channel 9; Touch5：Touch Input Channel 5;
14	PB4	I/O	8	GPIO	PWM3：Timer3 PWM Output; AMUX0L：Simulator Channel0 Left; SPI1CLKA：SPI1 Clock(A); SD0CMDB：SD0 Command(B); ADC8：ADC Input Channel 8; SPI0_DAT2AB(2)：SPI0 Data2(AB); Touch4：Touch Input Channel 4;
15	PB0	I/O	8	GPIO	UART1TXA：Uart1 Data Out(A); SPI2CLKA：SPI2 Clock(A); ADC6：ADC Input Channel 6; Touch0：Touch Input Channel 0;
16	VBAT	P	/	LDO Power	
17	PR1	I/O	10	RTCIO1 (output 0V)	RESET1： ADC12：ADC Input Channel 12;
18	PR2	I/O	10	RTCIO2 (pull up)	RESET2： ADC12：ADC Input Channel 12;
19	BT_AVDD	P	/	BT Power 1.3v	
20	BT_RF	P	/		
21	FMIP	I	/		
22	VSSIO	P	/	Ground	
23	BT_OSCI	I	/	BT OSC In	
24	BT_OSCO	O	/	BT OSC Out	

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2、Electrical Characteristics

2.1 PMU Characteristics

Table 2-1

Symbol	Parameter	Min	Typ	Max	Unit	Test Conditions
VBAT	Voltage Input	2.2	3.7	5.5	V	
$V_{3.3}$	Voltage output	_	3.3	_	V	LDO5V = 5V, 100mA loading
$V_{1.2}$		_	1.2	_	V	LDO5V = 5V, 50mA loading
$V_{1.3}$	Voltage output		1.3		V	LDO5V=5V, 100mA loading
V_{DACVDD}	DAC Voltage	_	3.1	_	V	LDO5V = 5V, 10mA loading
$I_{L3.3}$	Loading current	_	_	150	mA	LDO5V = 5V

2.2 IO Input/Output Electrical Logical Characteristics

Table 2-2

IO input characteristics						
Symbol	**Parameter**	**Min**	**Typ**	**Max**	**Unit**	**Test Conditions**
V_{IL}	Low-Level Input Voltage	-0.3	_	0.3* VDDIO	V	VDDIO = 3.3V
V_{IH}	High-Level Input Voltage	0.7* VDDIO	_	VDDIO+0.3	V	VDDIO = 3.3V
IO output characteristics						
V_{OL}	Low-Level Output Voltage	_	_	0.33	V	VDDIO = 3.3V
V_{OH}	High-Level Output Voltage	2.7	_	_	V	VDDIO = 3.3V

2.3 Internal Resistor Characteristics

Table 2-3

Port	General Output	High Drive	Internal Pull-Up Resistor	Internal Pull-Down Resistor	Comment
PA0 PA3 PA4 PC3~PC5	8mA	24mA	10K	10K	1、USBDM & USBDP default pull down 2、internal pull-up/pull-down resistance | accuracy ±20% 3、PR2 default pull up
PB0 PB4 PB5	4mA	8mA	10K	10K	
PR1 PR2	8mA	10mA	10K	10K	
USBDM USBDP	4mA	_	1.5K	15K	

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2.4 DAC Characteristics

Table 2-4

Parameter	Min	Typ	Max	Unit	Test Conditions
Frequency Response	20	_	20K	Hz	1KHz/0dB 10Kohm loading With A-Weighted Filter
THD+N	_	-69	_	dB	
S/N	_	95	_	dB	
Crosstalk	_	-80	_	dB	
Output Swing		1		Vrms	
Dynamic Range		90		dB	1KHz/-60dB 10Kohm loading With A-Weighted Filter
DAC Output Power	11		_	mW	32ohm loading

2.5 ADC Characteristics

Table 2-5

Parameter	Min	Typ	Max	Unit	Test Conditions
Dynamic Range		85		dB	1KHz/-60dB 10Kohm loading With A-Weighted Filter
S/N	_	90	_	dB	1KHz/-60dB 10Kohm loading With A-Weighted Filter
THD+N	_	-72	_	dB	
Crosstalk	_	-80	_	dB	

2.6 BT Characteristics

2.6.1 Transmitter

Basic Data Rate **Table 2-6**

Parameter		Min	Typ	Max	Unit	Test Conditions
RF Transmit Power			0	4	dBm	25℃, Power Supply Voltage=5V 2441MHz
RF Power Control Range			20		dB	
20dB Bandwidth			950		KHz	
Adjacent Channel Transmit Power	+2MHz		-40		dBm	
	-2MHz		-38		dBm	
	+3MHz		-44		dBm	
	-3MHz		-35		dBm	

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Enhanced Data Rate **Table 2-7**

Parameter		Min	Typ	Max	Unit	Test Conditions
Relative Power			1.2		dB	25℃, Power Supply Voltage=5V 2441MHz
π/4 DQPSK Modulation Accuracy	DEVM RMS		6		%	
	DEVM 99%		10		%	
	DEVM Peak		15		%	
Adjacent Channel Transmit Power	+2MHz		-40		dBm	
	-2MHz		-38		dBm	
	+3MHz		-44		dBm	
	-3MHz		-35		dBm	

2.6.2 Receiver

Basic Data Rate **Table 2-8**

Parameter		Min	Typ	Max	Unit	Test Conditions
Sensitivity			-89		dBm	25℃, Power Supply Voltage=5V 2441MHz
Co-channel Interference Rejection			-13		dB	
Adjacent Channel Interference Rejection	+1MHz		+5		dB	
	-1MHz		+2		dB	
	+2MHz		+37		dB	
	-2MHz		+36		dB	
	+3MHz		+40		dB	
	-3MHz		+35		dB	

Enhanced Data Rate **Table 2-9**

Parameter		Min	Typ	Max	Unit	Test Conditions
Sensitivity			-89		dBm	25℃, Power Supply Voltage=5V 2441MHz
Co-channel Interference Rejection			-13		dB	
Adjacent Channel Interference Rejection	+1MHz		+5		dB	
	-1MHz		+2		dB	
	+2MHz		+37		dB	
	-2MHz		+36		dB	
	+3MHz		+40		dB	
	-3MHz		+35		dB	

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2.7 FM Receiver Characteristics

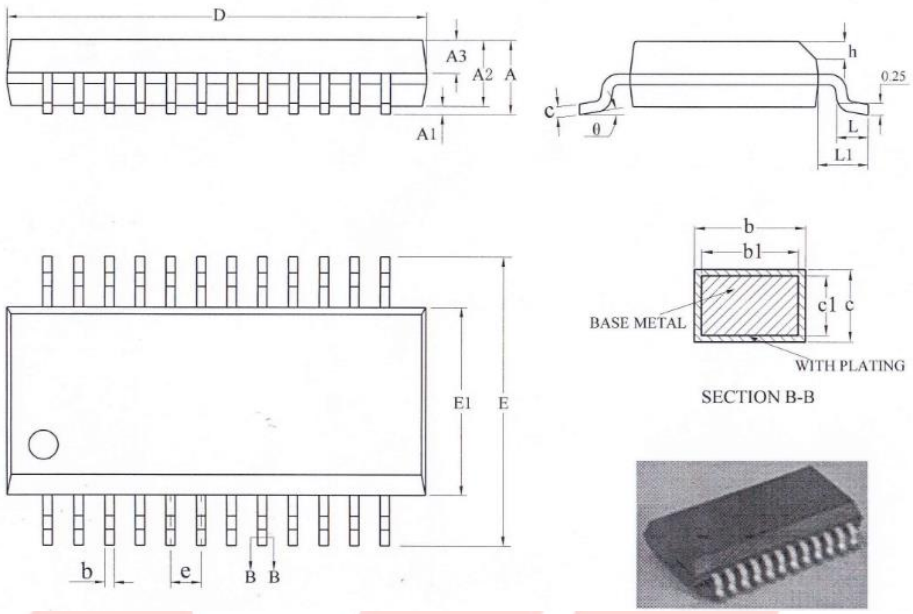
Table 2-10

Parameter	Min	Typ	Max	Unit	Test Conditions
Input Frequency	76		108	MHz	
Usable Sensitivity	3	4	8	dBμV EMF	(S+N)/N=26dB
Adjacent Channel Selectivity		48		dB	± 200kHz
IIP3		88		dbμV EMF	Δf1=200 kHz, Δf2=400 kHz
Audio Output Voltage	0		3	V	Empty load
Audio Frequency Response	20		20k	Hz	DAC test
Audio (S+N)/N		52		dB	
Stereo Separation		40		dB	
Audio Total Harmonic Distortion (THD)		0.4		%	

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3、Package Information

3.1 QSOP24



SYMBOL	MILLIMETER		
	MIN	NOM	MAX
A	—	—	1.75
A1	0.10	0.15	0.25
A2	1.30	1.40	1.50
A3	0.60	0.65	0.70
b	0.23	—	0.31
b1	0.22	0.25	0.28
c	0.20	—	0.24
c1	0.19	0.20	0.21
D	8.55	8.65	8.75
E	5.80	6.00	6.20
E1	3.80	3.90	4.00
e	0.635BSC		
h	0.30	—	0.50
L	0.50	—	0.80
L1	1.05REF		
θ	0	—	8°

Figure 3-1. AC6925E_QSOP24 Package

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4、Revision History

Date	Revision	Description
2018.09.05	V1.0	Initial Release

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